Capstone Project: Mortality Prediction in Patients with AKI

DSA 8401: Applied Machine Learning

Master in Data Science and Analytics

Strathmore University

@iLabAfrica Centre

As a capstone project, we seek to apply the knowledge acquired during the development of this course. Each team of 4 members is expected to train a machine learning model capable of predicting in-hospital mortality in the ICU for patients with acute kidney injury (AKI). Each group must make a 10-minute presentation that demonstrates the understanding and management of the knowledge obtained throughout the module. During the presentation, the group members are expected to present the evaluated models, the methodologies implemented, the challenges faced and the solutions adopted until reaching the final version of the learning model. The performance results of the final model should not be revealed in the presentation.

To evaluate the real performance of the model, the original data set has been divided into two sets: one for the generation of the models, available on the virtual campus, and another reserved for the final evaluation by the teachers. In this way, the aim is to simulate a realistic scenario where the performance of the model can be measured with data not previously seen by the machine learning system.

1 Objectives and Data Description

The objective is to predict in-hospital mortality in the ICU of a retrospective cohort of patients with AKI. The data comes from the MIMIC-III database. Each registry includes a set of variables that summarize the clinical trajectory

Variable	Description
Age	Patient's age
Sex	Patient's sex
Days in ICU	Days in IC
BP	Blood Pressure
HR	Heart Rate
Temp	Temperature
WBC	White Blood Count
PaO2	Partial pressure of Oxigen
FiO2	Fraction inhaled of Oxigen
Pot	Potasium
Bic	Bicarbonate
Na	Sodium
BUN	Blood Ureic Nitrogen
Bilirubin	Bilirubin
GCS	Glasgow Coma Scale
IMH	In-hospital mortality

Table 1: Description of the variables in the dataset

of the patients during their stay in the ICU. This information is detailed in the table shown below. The target variable, in-hospital mortality (IHM), is defined as a binary variable where the value '1' indicates the death of the patient in the ICU.

2 Evaluation

The evaluation of the project consists of two main components: the presentation (80%) and the validation of the final model (20%) using the reserved data. The submission of a written report is not required.

During the presentation of the project, the active participation of all team members is mandatory. The score for this component is divided equally between the overall quality of the proposal and the individual evaluation of each team member's contributions, as well as their ability to answer questions related to the proposal.

The remaining 20% corresponds to the evaluation of the models with the reserved data. The projects are expected to be developed on the Google Colab platform. It is important to note that the models will be evaluated using a data set exclusively accessible to teachers. The only material that must be delivered is a PDF file uploaded to the virtual campus, containing the link to the Jupyter notebook where the final model was developed. Following the procedures specified in this notebook, validation will be carried out with the training data and later with the test data. The last slide must describe the tasks carry on by every member. It's not allowed to assign one task to more than one member.

The metrics that will be used for the evaluation are the area under the ROC curve and the F-measure. To calculate the most optimal F-measure, each team must select and provide the probability threshold that will be applied to the reserved data.